The joint finite-time memory kernel

Charge–energy regression, exact static removal, and mixed spectra

Alan Day Chengheng*

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Abstract

We complete a convergent construction of the charged and neutral finite-time memory covariance in the common-coupling complex DSSYK model. A joint charge source identifies the conserved fluctuations left by internal Gaussian contractions. Subtracting their explicit charge–energy regression gives a strong reduction to pure oriented Wick sources in the same disorder sample. The residual two-energy spectra are absolutely continuous, permitting the limit of exact samplewise energy pinching. Uniform ordinary-power tails and a subsequent Hermite resummation give the full AA, AB, BA, and BB kernel. The construction retains the previously nonzero mixed calibration. The bulk de Sitter interpretation remains a separate next problem.

1 Setting and the actual memory observable

Use standard CAR, overlapping complex Gaussian p-body couplings, the same original disorder in every charge sector, and normalized annealed full-Fock trace at $\beta = \mu = 0$. Take even N, $p \to \infty$, $\lambda_N = p^2/N \to \lambda > 0$, fixed J_s, $\hbar = 1$, and string time. Set

$$C = J_s^2/\lambda, \quad q = e^{-4\lambda}, \quad X_N = (Q - N/2)/p, \quad b_{N,m} = 2^{-N}\binom{N}{m}. \tag{1}$$

The exact memory field from the preceding reduction is

$$\mathsf{F}_{m,N}(s) = (1 - \mathcal{P}_H)[me^{-ihs/2}M_m(s)e^{-ihs/2}], \quad \mathcal{P}_H Y = \sum_e P_e Y P_e, \quad h = H_N|_m. \tag{2}$$

For $m > 0$, its finite-dimensional definition is

$$V\psi = m^{-1/2}\sum_i |i\rangle c_i \psi, \quad K = I_N \otimes H_{m-1}, \quad Q_\perp = 1 - VV^\dagger, \tag{3}$$

$$R = Q_\perp K V, \quad K_\perp = Q_\perp K Q_\perp, \quad M_m(s) = R^\dagger e^{isK_\perp} R. \tag{4}$$

The $m = 0$ field is zero. Pinching retains entire degenerate energy blocks. The established two-component reduction is $\mathsf{F}_N = A_N + B_N + o_{L^2}(1)$, uniformly on compact memory-time intervals, with both components uniformly bounded. The neutral component B_N is constructed in [4]; the missing charged and mixed limits are the subject here.

Write $DY = QY - \sum c_i^\dagger Y c_i$, and let $\mathcal{N}$ be coupling Gaussian number. The preceding oriented-transfer result [3] supplies the complete coefficients

$$G_{n,N} = [p\mathcal{N} - D - \lambda_N \mathcal{N}(\mathcal{N} - 1)]H_N^n. \tag{5}$$

All polynomial degrees and real times in the limiting statements are fixed before $N \to \infty$. We do not take a joint $\lambda \to 0$ limit.

*AI research assistant (OpenAI).

2 A charge source identifies the conserved component

For a complex charge source t in a fixed disk,

$$\operatorname{tr} e^{tX_N} = \cosh(t/(2p))^N \longrightarrow e^{t^2/(8\lambda)}, \qquad \pi_t = \frac{e^{t/p}}{1+e^{t/p}}. \tag{6}$$

The local product-state method is the chemical-potential construction of [1, Sec. 4.2], applied at a source of order $1/p$. The marked loss source is the new ingredient inherited from [3]. For $z^2 = 0$,

$$e^{-zD_i} c_i^{(\dagger)} = e^{-z/2} c_i^{(\dagger)}, \quad e^{-zD_i} n_i = (1-z) n_i, \quad e^{-zD_i}(1-n_i) = 1 - n_i + z n_i. \tag{7}$$

An internal Gaussian chord has one n and one $1-n$ singleton. Their normalized source product, repeated at p sites, is exactly

$$[1 + (e^{t/p} - 1)z]^p = 1 + p(e^{t/p} - 1)z \longrightarrow e^{tz}. \tag{8}$$

Open-leg source factors cancel exactly against e^{pdz}. A neutral size mark on a Wick subblock has only open legs and adds no charge term. For ℓ unmarked chords the remaining singleton correction is

$$[4\pi_t(1-\pi_t)]^{p\ell} = \cosh(t/(2p))^{-2p\ell} = 1 + O_T(p^{-1}). \tag{9}$$

This factor is discarded only after the source cancellation.

For two whole-field G insertions, let i_a count internal chords, O_a the chords with exactly one field endpoint, and $B = O_1 \cap O_2$. Put $k_a = -4\,\mathrm{cr}(O_a) - 2i_a$ and $K_D(B) = 2|B|(|B|-1) - 4\,\mathrm{cr}(B)$. Their joint tilted weight is

$$C^\ell q^{\mathrm{cr}\,\pi} e^{t^2/(8\lambda)} \left\{ \lambda[4i_1 i_2 + K_D(B)] + (\lambda k_1 + i_1 t)(\lambda k_2 + i_2 t) \right\}. \tag{10}$$

Equivalently, average over $X \sim \gamma_\lambda = N(0, 1/(4\lambda))$ the conditional single-source weights $\lambda(k_a + 4i_a X)$ and connected weight $\lambda K_D(B)$. The former reproduce the previously found $4\lambda i_1 i_2$ static covariance.

For completeness, the convergence is locally uniform in complex t. After singleton cancellation, each fixed coefficient is bounded by $K(1+R)^r e^{cR}$, where R is the total overlap of finitely many p-sets. Hypergeometric exponential moments are uniformly bounded; triple intersections vanish under this domination. Singleton denominators remain bounded away from zero. Cauchy's formula then gives every fixed X_N-power moment. Positive squared norms tilted at $t = \pm a$ control their charge tails. Thus the commuting scalar law is $\gamma_\lambda \otimes \rho_q$, with the required weighted uniform integrability; this does not assert independence of all dynamic fields.

3 Explicit joint regression and a strong common-copy reduction

Let $H_d^{(q,C)}$ be monic q-Hermite polynomials with

$$H_{d+1}^{(q,C)}(x) = xH_d^{(q,C)}(x) - C[d]_q H_{d-1}^{(q,C)}(x), \quad P_d = [d]_q!, \quad \|H_d^{(q,C)}\|_{\rho_q}^2 = C^d P_d, \tag{11}$$

and set $m_d = q\partial_q \log P_d$, $m_0 = m_1 = 0$. Define

$$x^n = \sum_d a_{nd} H_d^{(q,C)}(x), \qquad a_{n+1,d} = a_{n,d-1} + C[d+1]_q a_{n,d+1}, \quad a_{00} = 1. \tag{12}$$

Only $0 \le d \le n$ of matching parity occur, and $a_{nd} \ge 0$.

Proposition 1. *The limiting orthogonal regression onto the commuting charge/energy space is*

$$\boxed{g_n(X,H) = \sum_d a_{nd}[-4\lambda m_d - \lambda(n-d)(1-2X)]H_d^{(q,C)}(H).} \tag{13}$$

If $Z_{d,N} = \mathcal{W}_d(H_N^d)$ is highest coupling chaos and

$$J_{d,N} = (dp - D)Z_{d,N} - \kappa_d^- H_d^{(q,C)}(H_N), \tag{14}$$

$$\kappa_d^- = \lambda[d(d-1) - 4m_d], \qquad R_{n,N} = G_{n,N} - g_n(X_N, H_N), \tag{15}$$

then, with fixed left/right energy weights allowed,

$$\left\| R_{n,N} - \sum_d a_{nd} J_{d,N} \right\|_2 \longrightarrow 0. \tag{16}$$

Here $J_0 = J_1 = 0$; this J_d is a field, not the disorder label.

Proof. Fix an internal partial matching γ of the n-slot field, with i internal chords and $d = n - 2i$ open slots. Its weight is $C^i q^{c_\gamma}$, where c_γ includes internal–open crossings. The open endpoint permutations have weight $q^{\operatorname{inv}\sigma}$. Their conditional source is $\lambda[-4\operatorname{inv}\sigma - 2i + 4iX]$, whose average is $\lambda[-4m_d - 2i + 4iX]$. Summing internal matchings gives (13). Charge and energy polynomials are dense in $L^2(\gamma_\lambda \otimes \rho_q)$, so the calculated orthogonality is the stated projection.

After subtracting this regression, the conditional source is $-4\lambda(\operatorname{inv}\sigma - m_d)$, with connected weight $\lambda K_D(B)$. This is precisely the centered pure J_d rule; all i and X terms have canceled. Internal matchings leave only a_{nd}. The same argument computes the mixed R_n–J_e Gram terms in the same disorder sample. Expanding the squared norm of their difference proves (16), rather than only equality of two marginal laws. Fixed Gaussian-degree multiplication bounds permit energy weights. No loss operator is applied to an uncontrolled $Z_d - H_d^{(q,C)}(H_N)$ error. □

4 Pure residuals and the absence of extra static mass

With $v_d = (q\partial_q)^2 \log P_d$, the pure-source rule gives

$$\|J_d\|_2^2 = C^d P_d\{\lambda[2d(d-1) - 4m_d] + 16\lambda^2 v_d\}. \tag{17}$$

These are limiting norms. The Mahonian inversion bound implies

$$\|J_d\|_2 \le K(\sqrt{\lambda}\, d + \lambda d^2)\sqrt{C^d P_d}. \tag{18}$$

In $J_d^\dagger H^a J_e H^b$, fix all external partner positions. If there is no direct field bridge, $K_D(B) = 0$. The two open permutation sums are independent and centered, so this entire group cancels. Individual diagrams need not vanish.

Let $p_m = H_m^{(q,C)}/\sqrt{C^m P_m}$ and $M_{mn}^{de} = \lim \mathbb{E}\,\mathrm{tr}[J_d^\dagger p_m(H) J_e p_n(H)]$. External Hermite blocks remove their own internal pairs by the multiblock Wick identity [2]. Deleted chords belong to no marked incident set, so this cancellation preserves the marks. Each remaining field bridge crosses every energy bridge. For a fixed $D = d + e$, endpoint counting and the remaining permutation sum give the same estimate as in [4]:

$$|M_{mn}^{de}| \le K_D(m+n+1)^D q^{\max(0,\min(m,n)-D)} \mathbf{1}_{|m-n|\le D}. \tag{19}$$

Here the energy-bridge count is at least $\min(m,n) - D$, and $P_r/\sqrt{P_m P_n} \le (q;q)_\infty^{-1}$. Thus the coefficients are square summable over both external degrees.

Use the positive matrix measures

$$\nu_{ab,N}(A,B) = \mathbb{E}\,\mathrm{tr}[\mathcal{X}_{a,N}^\dagger P_H(B)\mathcal{X}_{b,N} P_H(A)]. \tag{20}$$

Fixed marked moments, positivity, and energy-moment uniform integrability give a weak limit. Its compact support follows explicitly from the fixed-mark bound and ordinary q-Gaussian moments:

$$\int E^{2m}\, d\nu_{dd} \le K_d R_H^{2m}, \quad \int (E')^{2m}\, d\nu_{dd} \le K_d R_H^{2m}, \quad R_H = 2\sqrt{C/(1-q)}. \tag{21}$$

The constants absorb the fixed field degrees and are independent of m. Positivity extends support to cross measures; compact moment determinacy gives uniqueness. Equation (19) constructs its density:

$$d\nu_{de} = f_{de}\, d\rho_q(E) d\rho_q(E'), \qquad f_{de} = \sum_{m,n\ge 0} M_{mn}^{de} p_n(E) p_m(E'). \tag{22}$$

The series converges in L^2. Since ρ_q is diffuse, the diagonal $\Delta = \{E = E'\}$ has zero mass. The same holds for R_n and its fixed energy multiples by (16). Portmanteau on this closed diagonal now gives

$$\limsup_N \|\mathcal{P}_H R_{n,N}\|_2^2 = \limsup_N \nu_{R_n R_n, N}(\Delta) \le 0. \tag{23}$$

Because $g_n(X_N, H_N)$ commutes with H_N exactly, this proves the actual static subtraction, including degeneracies. It exchanges neither an infinite-time average nor a growing-time limit with N.

5 Neutral descendants and the common matrix spectrum

For a self-contained specification of the combined field list, recall the neutral construction [4]. With $Z_j = \mathcal{W}_j(H^j)$, let $U_j = (jp - \mathcal{L})Z_j$, where $\mathcal{L}$ is half Clifford size, and

$$T_{v,w} = U_v \diamond Z_w, \tag{24}$$

$$r_j = 4\lambda C \sum_{1\le v<j} q^v T_{v,j-1-v}, \tag{25}$$

$$\eta_{j+1} = H\eta_j - C[j]_q \eta_{j-1} - r_j, \qquad \eta_0 = \eta_1 = \eta_2 = 0, \tag{26}$$

$$V_j = 4\lambda q^j U_j + (q^j - q^{\mathcal{N}})\eta_j, \qquad N_j = V_j - v_j(H), \quad v_j = \operatorname{Reg} V_j. \tag{27}$$

The diamond is coupling-Gaussian Wick multiplication, with ordered matrix coefficients. The entire $v + w$ source forbids internal pairing, while the physical size mark acts only on its v subblock. The whole-field internal-pair count selects the degree on which $q^{\mathcal{N}}$ acts. An adjoint reflects every interval.

The scalar regressions are also finite recurrences. Put $\kappa_v^L = \lambda[v^2 - 4m_v]$ and $g_j^\eta = \operatorname{Reg}\eta_j$. Then

$$g_{j+1}^\eta(x) = x g_j^\eta(x) - C[j]_q g_{j-1}^\eta(x) - 4\lambda C\Big(\sum_{v<j} q^v \kappa_v^L\Big) H_{j-1}^{(q,C)}(x), \tag{28}$$

$$v_j(x) = 4\lambda q^j \kappa_j^L H_j^{(q,C)}(x) + \sum_d (q^j - q^d) g_{j,d}^\eta H_d^{(q,C)}(x), \tag{29}$$

where $g_j^\eta = \sum_d g_{j,d}^\eta H_d^{(q,C)}$ and the first three g_j^η vanish. Terms at $v = 0$ are zero.

All ν_{ab} for $\mathcal{X}_a \in \{J_d, N_j\}$ are now determined by a single algorithm: expand these finite recurrences, impose their Wick restrictions, insert the local oriented/size marks, and sum common-disorder pairings. For a mixed pair the connected coefficient is

$$K_{DL} = |B_{DL}|(|B_{DL}| - 1) - 4\,\mathrm{cr}(B_{DL}), \qquad B_{DL} = O_D \cap I_L. \tag{30}$$

Here I_L is incident to the smaller neutral marked subblock; it may contain chords internal to the larger neutral field. It must not be replaced by that larger field's open set. Neutral–neutral weights and their regression use the established size rule.

The fixed neutral diagonal densities are already L^2. The combined matrix measure is positive, so its mixed density satisfies $|f_{ab}|^2 \le f_{aa} f_{bb}$. Cauchy–Schwarz gives mixed L^2 integrability as well. Thus (22), with the computed mixed Hermite moments, constructs the entire fixed-field matrix. No mixed block is set to zero by an independence assumption.

6 Finite-time passage before Hermite resummation

The actual charged transfer is controlled by the already proved ordinary coefficients

$$K_{n,N} = G_{n+2,N} - \{H_N, G_{n+1,N}\} + H_N G_{n,N} H_N, \tag{31}$$

$$\|K_{n,N}\|_2 \le K'(n+3)^4 L^{n+2}\sqrt{n!}. \tag{32}$$

On a compact real-time interval, the charged field is strongly approximated by

$$(1-\mathcal{P}_H)e^{-iH_N s/2}\sum_{n\ge 0}\frac{(is)^n}{n!}K_{n,N}e^{-iH_N s/2}. \tag{33}$$

First truncate this series at finite N. Subtract the exactly commuting polynomial $k_n = g_{n+2} - 2Hg_{n+1} + H^2 g_n$, then use (23). The limiting residual coefficient is

$$L_n = R_{n+2} - \{H, R_{n+1}\} + HR_nH. \tag{34}$$

The joint limiting regression is an orthogonal H-bimodule projection, so $\|L_n\|_2 \le \|K_n\|_2$. This and the finite-N projected tail allow the cutoff to tend to infinity. We never require the limiting g_n to be a finite-N conditional expectation.

Only now work in the limiting Hilbert space. Nonnegativity of a_{nd}, orthogonality of $H_d^{(q,C)}$, and compact energy support yield

$$\sum_d a_{nd}\sqrt{C^d P_d} \le \sqrt{n+1}\|H^n\|_{\rho_q} \le \sqrt{n+1}R_H^n. \tag{35}$$

Together with (18), this proves absolute summability of the double series and its required derivatives:

$$\sum_{n,d}\frac{S^n}{n!}a_{nd}\|J_d\|_2 \le K\sum_n \frac{(SR_H)^n}{n!}(n+1)^{5/2} < \infty. \tag{36}$$

Define the computable time coefficients

$$c_d(s) = \frac{\int e^{isE}H_d^{(q,C)}(E)\,d\rho_q(E)}{C^d P_d} = \sum_n \frac{(is)^n}{n!}a_{nd}. \tag{37}$$

Then $\sum_n (is)^n R_n/n! = \sum_d c_d(s)J_d$. Left/right energy actions are bounded by R_H, so the charged force amplitude is

$$D_d(s;E,E') = -c_d''(s) + i(E+E')c_d'(s) + EE'c_d(s). \tag{38}$$

The neutral amplitude is $+2Cc_j(s)$, with its previously proved summable descendant tail. Both carry the same centering phase $e^{-is(E+E')/2}$. This resummation includes all lower coupling chaoses rather than dropping them from the original exponential.

7 The complete joint covariance

Use the field list $\mathcal{X}_a \in \{J_d\ (d \geq 2), N_j\ (j \geq 1)\}$, the common positive matrix spectrum (20), and amplitudes

$$f_{J_d}(s; E, E') = D_d(s; E, E'), \qquad f_{N_j}(s; E, E') = 2Cc_j(s). \tag{39}$$

Our convention is E = input/column and E' = output/row.

Theorem 1. *For fixed real s, u, T, the actual full-trace memory covariance has the convergent representation*

$$\begin{aligned} &\lim_N \mathbb{E}\,\mathrm{tr}[\mathsf{F}_N(s)^\dagger e^{iTH_N}\mathsf{F}_N(u)e^{-iTH_N}] \\ &\quad = \sum_{a,b}\int e^{\frac{i}{2}(s-u)(E+E')+iT(E'-E)}\overline{f_a(s; E, E')}f_b(u; E, E')\,d\nu_{ab}(E, E'). \end{aligned} \tag{40}$$

All AA, AB, BA, and BB terms are included. Fixed-field densities are $L^2(\rho_q \otimes \rho_q)$; the full finite-time spectral matrix is absolutely continuous and has no diagonal static mass.

Proof. For finite field cutoffs the common moment construction determines the positive spectral matrix; exact static removal follows from (23). Weighted transfer, the ordinary tails, and the neutral reduction identify this with the actual field's covariance. Section 6 then resums the charged component. Spectral Cauchy–Schwarz bounds the variation of a cross measure by the product of the corresponding vector norms. The charged and neutral summable tails therefore make the double sum absolutely convergent in spectral total variation, uniformly for s, u in compact intervals. Multiplication by the external real-time phase does not change these tail bounds. Absolute continuity is preserved under total-variation limits, completing the argument. □

The positive matrix construction gives a nonnegative equal-time norm and the conjugation relation $\mathcal{K}(s, u; T)^* = \mathcal{K}(u, s; -T)$. The infinite density is only asserted to be L^1; the fixed-field L^2 constants need not be uniform as the field degree grows. Likewise, uniformity over all real T applies to the series tails, not to convergence of the original finite-N spectra.

Equation (40) is a constructive full kernel, not a finite Taylor fit. Every matrix coefficient is given by finite common-J pairing sums, its energy dependence by a convergent Hermite expansion, and its time dependence by (38). A short special-function closed form is not required for this construction. Extracting the bulk particle content and gravitational interaction dictionary from this spectrum is the next physical task.

8 Low-order contact with the earlier spectra

The time coefficients immediately give

$$A_0 = J_2, \tag{41}$$
$$A_1 = J_3 - \tfrac{3}{2}\{H, J_2\}, \tag{42}$$
$$B_1 = 2CN_1, \qquad B_2 = 2C[N_2 - \{H, N_1\}], \tag{43}$$

where $A(s) = A_0 + isA_1 + \cdots$ and $B(s) = isB_1 - s^2B_2/2 + \cdots$. For G_2 the joint regression is

$$g_2 = -\frac{4\lambda q}{1+q} H_2^{(q,C)}(H) - 2\lambda C + 4\lambda CX. \tag{44}$$

The last term carries variance $4\lambda C^2$. Removing the energy regression alone leaves that term; removing both gives

$$\nu_{A_0A_0} = w\mu_{q^2}, \qquad w = 4C^2\lambda + \frac{16C^2\lambda^2 q}{1+q}, \tag{45}$$

in agreement with the previously identified Y_- spectrum [5]. The new common-source moments also recover the nonzero mixed channel

$$\nu_{A_0B_2} = d_0(w+\eta)\mu_{q^2}, \quad d_0 = 4\lambda C q^2, \quad \eta = -4C^2\lambda q + \frac{16C^2\lambda^2 q}{1+q}. \tag{46}$$

Here μ_r is the probability q-Mehler measure characterized by

$$\int H_a^{(q,C)}(E) H_b^{(q,C)}(E')\, d\mu_r = \delta_{ab} r^a C^a P_a. \tag{47}$$

For physical $\lambda > 0$, $d_0(w+\eta) > 0$. The A_0–B_1 channel vanishes. Independent saved checks test (46) through total external energy degree four, while the general all-degree measure is supplied by the construction.

The previous named decomposition $A_1 = R + a\{H, Y_-\}/2$, with $a = -(1-q)^2/(1+q^2)$, can now be compared through the explicit target

$$J_3 = R + \frac{2(1+q+q^2)}{1+q^2}\,\frac{\{H, Y_-\}}{2}. \tag{48}$$

We retain this as a remaining naming/calibration cross-check, not an assumption in the full-kernel proof.

The present result closes the finite-time joint memory calculation in this stated model and limit. It does not yet establish its de Sitter mass/position interpretation, a finite-temperature extension, a uniform small-λ limit, or finite-G gravitational four-point dynamics. The next stage should use the constructed spectral matrix to test the physical dictionary rather than repeat its existence proof.

9 Validation and provenance

Three new scientific outputs accompany the proof; each refuses to overwrite an existing result. The stage evidence records parameters, dependency versions, predecessor hashes, build logs, and final page renders. No predecessor scientific main was rerun.

- Joint regression: 68 charge/energy projection checks, 39 two-energy Gram layouts with 117 equalities, 30 no-bridge grouped cancellations (28 had nonzero uncentered values), 44 mixed-neutral layouts, and 16 comparisons with the sealed neutral engine. There are 18 nonzero A_0B_2 Mehler tests and 12 zero A_0B_1 tests. The recorded maximum is ten raw slots, G_n through $n = 5$ and neutral V_j through $j = 4$. Rational q, λ are formal independent variables, not asserted to satisfy $q = e^{-4\lambda}$.

- Charge tilt: complete finite CAR is compared with the local charge-generating polynomial: 19,502 support/pairing/layout cases and 315,192 coefficients, including smaller neutral mark intervals and whole-field internal contractions. There are 16 exact singleton and 16 collision-ratio checks, 216 complex scalar tilts, and 20 full-charge G_2 tilts compared at 70 and 110 digits. Finite errors are retained; these high-precision comparisons are not interval certificates.

- Time coefficients: 90 rational ordinary/Hermite identities, 1782 centered-force coefficient identities, and 405 physical-parameter quadrature coefficients for c_d, c_d', c_d''. Independent Jacobi sizes 48 and 72 differ by at most 1.281×10^{-14}. The recorded analytic bound controls Taylor/quadrature truncation only; floating roundoff is not interval-certified. These calculations evaluate the limiting coefficient functions, not finite-N dynamics.

The tests check the claimed interfaces. The arbitrary-degree regression, static removal, and time-limit interchange rely on the proofs above and the cited predecessor bounds. All outputs remain local; this stage makes no Git publication or external issue reply.

References

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[3] A. Day and Chengheng, *Oriented sources for the charged memory field*, local project stage `susskind-memory-oriented-transfer` (2026).

[4] A. Day and Chengheng, *A constructive neutral memory kernel*, local project stage `susskind-memory-wick-descendants` (2026).

[5] A. Day and Chengheng, *Two-component finite-memory reduction*, local project stage `susskind-memory-two-component` (2026); see also the preceding projected-memory and uniform-bound stages.